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Webinar: #2

PROBLEM 1

Consider the following linear optimization problem:

$$\begin{aligned}
\min_{x_1, x_2} \quad & 3x_1 + 3x_2 \\
\text{subject to} \quad & 2x_1 + x_2 \leq 3, \\
& x_1 + x_2 \leq 1, \\
& 2x_1 + 3x_2 \leq 5.
\end{aligned}$$

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(a) Rewrite this optimization problem in the following linear programming form learned in MFML course:

$$\min_{x \in \mathbb{R}^2} \quad c^T x$$
$$\text{subject to} \quad Ax \leq b$$

Mention clearly $A \in \mathbb{R}^{3\times 2}$, $c \in \mathbb{R}^2$ and $b \in \mathbb{R}^3$ in your solution.

(b) Derive the Lagrangian of the reformulated optimization problem. Derive the minimum of Lagrangian with respect to x.

(c) Derive the dual function $D(\lambda_1, \lambda_2, \lambda_3)$ where $\lambda_1, \lambda_2, \lambda_3$ are the Lagrange multipliers.

(d) Derive the dual optimization problem in terms of dual variables $\lambda_1, \lambda_2, \lambda_3$

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SOLUTION

(a) The given problem is:

$$\min \; 3x_1 + 3x_2$$

subject to

$$\begin{aligned} 2x_1 + x_2 &\leq 3, \\ x_1 + x_2 &\leq 1, \\ 2x_1 + 3x_2 &\leq 5. \end{aligned}$$

This can be written in the form $\min c^T x$ subject to $Ax \leq b$ as:

$$c = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 1 \\ 5 \end{bmatrix}.$$

(b) The Lagrangian is:

$$L(x, \lambda) = c^T x + \lambda^T (Ax - b),$$

where $\lambda = (\lambda_1, \lambda_2, \lambda_3)^T \geq 0$.

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SOLUTION

(c) The dual function is:

$$D(\lambda) = -(3\lambda_1 + \lambda_2 + 5\lambda_3),$$

(d) The dual optimization problem is:

$$\begin{aligned} \max_{\lambda_1,\lambda_2,\lambda_3} \quad & -(3\lambda_1 + \lambda_2 + 5\lambda_3) \\ \text{subject to} \quad & 2\lambda_1 + \lambda_2 + 2\lambda_3 = -3, \\ & \lambda_1 + \lambda_2 + 3\lambda_3 = -3, \\ & \lambda_1, \lambda_2, \lambda_3 \geq 0. \end{aligned}$$

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PROBLEM 2

Consider the following function

$$f(x) = e^{0.5x} + x^2 - 3x.$$

(a) Verify whether the function $f(x)$ is convex on the interval $x \in [-2, 4]$.

(b) Apply the gradient descent algorithm to approach the minimum of $f(x)$. Take an initial guess $x_0 = 0.5$ and learning rate 0.1. Perform **five iterations** of the gradient descent method and record the successive values of x_k and $f(x_k)$.

SOLUTION

Given:

$$f(x) = e^{0.5x} + x^2 - 3x$$

(a) To check convexity, compute the second derivative.

$$f'(x) = 0.5e^{0.5x} + 2x - 3$$

$$f''(x) = 0.25e^{0.5x} + 2$$

Since $e^{0.5x} > 0$ for all x, we have

$$f''(x) = 0.25e^{0.5x} + 2 > 0 \quad \forall x \in [-2, 4].$$

Hence, $f(x)$ is **strictly convex** on $[-2, 4]$. (Kindly not that this is a sufficient condition and not necessary condition.)

SOLUTION

(b)

$$x_{k+1} = x_k - \alpha f'(x_k), \quad \alpha = 0.1.$$

$$f'(x) = 0.5e^{0.5x} + 2x - 3.$$

Starting with $x_0 = 0.5$:

$$f'(0.5) \approx -1.358, \quad x_1 = 0.5 + 0.1358 = 0.6358, \quad f(x_1) = -0.1289$$

$$f'(0.6358) \approx -1.0413, \quad x_2 \approx 0.7399, \quad f(x_2) = -0.2246$$

$$f'(0.7399) \approx -0.7964, \quad x_3 \approx 0.8195, \quad f(x_3) = -0.2805$$

$$f'(0.8195) \approx -0.6078, \quad x_4 \approx 0.8803, \quad (x_4) = -0.313$$

$$f'(0.8803) \approx -0.4629, \quad x_5 \approx 0.9266, \quad f(x_5) = -0.3319$$

PROBLEM 3

Consider the data given below.

5	3	6	2	8	7
2.5	2.2	3.0	1.8	3.4	3.1

(a) Standardize the two variables (rows) so that each has zero mean and unit variance. Display the standardized data matrix.

(b) Compute the covariance matrix of the standardized data and determine its eigenvalues and eigenvectors. Interpret the direction of the first principal component.

(c) Project the original (unstandardized) data onto the first principal component to obtain the one-dimensional representation of the six observations. (2+2+2 Marks)

SOLUTION

(a) Mean of first row:

$$\mu_1 = \frac{5+3+6+2+8+7}{6} = 5.167$$

Mean of second row:

$$\mu_2 = \frac{2.5+2.2+3.0+1.8+3.4+3.1}{6} = 2.667$$

Standard deviations:

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$$\sigma_1 \approx 2.115, \quad \sigma_2 \approx 0.553$$

Standardized data:

$$Y = \frac{X-\mu}{\sigma}$$

$$Y \approx \begin{bmatrix} -0.079 & -1.025 & 0.394 & -1.497 & 1.339 & 0.867 \\ -0.302 & -0.844 & 0.602 & -1.568 & 1.325 & 0.783 \end{bmatrix}$$

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(b)

$$S = \frac{1}{n} YY^T$$

$$S = \begin{bmatrix} 1 & 0.988 \\ 0.988 & 1 \end{bmatrix}$$

Eigenvalues:

$$\lambda_1 \approx 1.988, \quad \lambda_2 \approx 0.012$$

Eigenvectors:

$$b_1 \approx \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b_2 \approx \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

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(c) First principal component:

$$b_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Projection of each observation:

$$z_k = b_1^T x_k$$

$$z_1 = \frac{1}{\sqrt{2}}(5 + 2.5) = 5.303$$
$$z_2 = \frac{1}{\sqrt{2}}(3 + 2.2) = 3.677$$
$$z_3 = \frac{1}{\sqrt{2}}(6 + 3.0) = 6.364$$
$$z_4 = \frac{1}{\sqrt{2}}(2 + 1.8) = 2.687$$
$$z_5 = \frac{1}{\sqrt{2}}(8 + 3.4) = 8.061$$
$$z_6 = \frac{1}{\sqrt{2}}(7 + 3.1) = 7.142$$

Thus, the one-dimensional representation is:

$$\{5.303,\ 3.677,\ 6.364,\ 2.687,\ 8.061,\ 7.142\}$$

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PROBLEM 4

Consider the following dataset.

x	y
1	5
2	7

To estimate $\hat{y}$, assume the linear model $\hat{y} = xw$.

(a) Write the Mean Squared Error loss function $L(w)$ for this dataset. (3 Marks)

(b) Perform two iterations of the AdaGrad update

$$w_{i+1} = w_i - \frac{\alpha}{\sqrt{A_i + \epsilon}} \left.\frac{\partial L}{\partial w}\right|_{w=w_i},$$

using the initial values $w_0 = 0$, $A_0 = 0$, learning rate $\alpha = 0.2$, and $\epsilon = 10^{-8}$. (3 Marks)

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Solution

(a)

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(x_i w - y_i)^2$$

Here $n = 2$, so

$$L(w) = \frac{1}{2}\left[(w-5)^2 + (2w-7)^2\right]$$

Gradient:

$$\frac{dL}{dw} = \frac{1}{2}\left[2(w-5) + 4(2w-7)\right] = (w-5) + 2(2w-7)$$

$$\frac{dL}{dw} = 5w - 19$$

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SOLUTION

(b) Given:

$$w_0 = 0, \quad A_0 = 0, \quad \alpha = 0.2, \quad \epsilon = 10^{-8}$$

Let

$$g_i = \frac{dL}{dw}\Big|_{w=w_i}$$

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Solution

Iteration 1:

$$g_0 = 5(0) - 19 = -19$$

$$A_1 = 0 + (-19)^2 = 361$$

$$w_1 = 0 - \frac{0.2}{\sqrt{361 + 10^{-8}}}(-19) = 0.2$$

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Solution

Iteration 2:

$$g_1 = 5(0.2) - 19 = -18$$

$$A_2 = 361 + (-18)^2 = 685$$

$$w_2 = 0.2 - \frac{0.2}{\sqrt{685 + 10^{-8}}}(-18) = 0.338$$

Final Values:

$$w_1 \approx 0.2, \quad w_2 \approx 0.338$$

PROBLEM 5

Assume that you are given the matrix A as below:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

You are asked to use an iterative technique (e.g., power method) to obtain the eigenvector corresponding to the dominant eigenvalue. Your start with an initial value of

$$x = [0, 0.5, 1]$$

for the iterative technique. To which eigenvalue and eigenvector will the algorithm likely converge? Give reasons for your answer.

$$X^{(1)} = AX_0 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 0 \\ 0.5 \\ 1 \end{bmatrix} = \lambda_1 X_1$$

$$X^{(2)} = AX_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 0 \\ 0.5 \\ 1 \end{bmatrix} = \lambda_2 X_2$$

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Here the method converges in second iteration.

$$\text{The largest eigenvalue } \lambda = 4$$

and the corresponding eigenvector

$$X = \begin{bmatrix} 0 \\ 0.5 \\ 1 \end{bmatrix}$$

The converged eigenvalue is the same as the given initial vector. The given matrix is symmetric positive definite with the dominant eigenvalue repeated. Thus, any vector in the eigenspace of the dominant eigenvalue is a valid eigenvector.

PROBLEM 6

Teacher asked the students to conduct PCA on training dataset. He obtained eigenvalues of covariance matrix as

$$0.014, \; 0.016, \; 1, \; 3.5, \; 6.8, \; 12$$

and gave to the students.

(i) He asked the students to find the dimension of each training sample. Students said it is impossible to find given the insufficient information. Prove or disprove the claim made by the students. (1 mark)

(ii) He further asked the students to find out the minimum number of principal components to retain after dimension reduction by PCA for the following cases:

  (a) If we want 95% of the variance to be retained.
  (b) If we want 99% of the variance to be retained.

  Students said in both the cases we have to keep atleast 4 components. Prove or disprove the claim made by student. (3 marks)

(iii) Teacher decided to keep the following two principal components:

$$\begin{pmatrix} 0.115 \\ 0.106 \\ 0.118 \\ 0.082 \\ 0.013 \\ 0.023 \end{pmatrix} \text{ and } \begin{pmatrix} 0.205 \\ 0.215 \\ 0.315 \\ 0.428 \\ 0.238 \\ 0.034 \end{pmatrix}.$$

Given a normalized training example

$$\begin{pmatrix} -3 \\ -2 \\ 2 \\ 1 \\ 2 \\ 4 \end{pmatrix},$$

he asked the students to find the result of PCA on this training example. Students said it is not possible to apply. Prove or disprove students' claim. (2 marks)

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(i) The dimension of each training sample is 1×6 as the covariance matrix has 6 eigenvalues. This means each training sample has 6 features.

(ii) Total variance $= \sum$ of all eigenvalues $=$ 23.330.If we want to retain 95% variance, then we should keep the top 3 eigenvalues:

$$\frac{(12 + 6.8 + 3.5) \times 100}{23.330} = 95.7$$

If we want to retain 99% variance, then we should keep the top 4 eigenvalues:

$$\frac{(12 + 6.8 + 3.5 + 1) \times 100}{23.330} = 99.87$$

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(iii) After PCA, the result will be:

$$\begin{bmatrix}-3 & -2 & 2 & 1 & 2 & 4\end{bmatrix}\begin{pmatrix}0.115 & 0.205\\0.106 & 0.215\\0.118 & 0.315\\0.082 & 0.428\\0.013 & 0.238\\0.023 & 0.034\end{pmatrix} = \begin{bmatrix}-0.1210 & 0.6250\end{bmatrix}$$

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